Code No: R163205A **R16** SET - 1

III B. Tech II Semester Supplementary Examinations, April - 2021

ARTIFICIAL INTELLIGENCE

(Common to Computer Science and Engineering, Information Technology)

Time: 3 hours Max. Marks: 70

Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
2. Answer **ALL** the question in **Part-A**
3. Answer any **FOUR** Questions from **Part-B**

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PART –A **(14 Marks)**

1. a) What are the current trends in AI? [2M]
   b) Define state-space search. [2M]
   c) What is axiomatic system? [2M]
   d) Define script structure. [3M]
   e) What are the applications of expert systems? [3M]
   f) Difference between Fuzzy set and Fuzzy. [2M]

PART –B **(56 Marks)**

2. a) What are the different steps followed in developing AI-languages? Explain. [7M]
   b) Explain about Artificial intelligence systems. [7M]

3. Describe with necessary diagrams, a suitable state space representation for 8 puzzle problem and explain how the problem can be solved by state space search? Show how heuristic can improve the efficiency of search? [14M]

4. a) Explain about axiomatic system. [7M]
   b) Consider the following facts: [7M]
   (i) John likes anyone who likes playing tennis and football
   (ii) A is brother of B if A is a male, A has father F and mother M and B has the same mother and father as A does. Translate these facts into formulae in predicate logic and prepositional logic.

5. a) What are the approaches to knowledge representation? Discuss. [7M]
   b) Explain in detail about script structure. [7M]

6. a) Write about various shells and tools used in expert system in detail. [7M]
   b) Explain in brief about Rule based expert systems. [7M]

7. a) Discuss about fuzzy set operations. [7M]
   b) Discuss about inference rules for fuzzy propositions. [7M]

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